

# Project Optimisation Techniques

## 2025-2026

### Brewery Plat-Eau

## 1 Brewery Plat-Eau

In this project the goal is to study the functioning of Brewery Flat-Eau and to solve the subsequent optimization problems. A little warning for the thirsty student, this assignment is a simplification of reality regarding recipes, formulas, quantities, and so on. If you plan to make your own beer brewing, take a real manual.

### 1.1 Introduction to the art of brewing

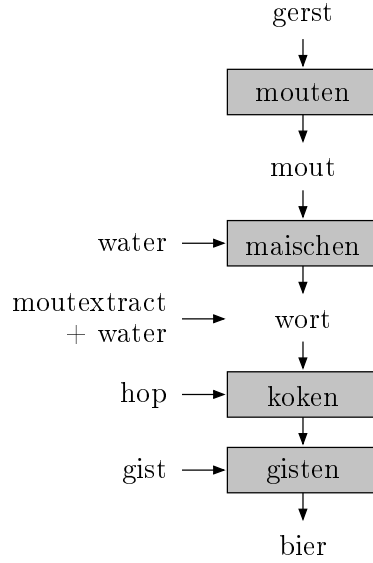
The four main ingredients of beer are: barley (Dutch: *gerst*), water, hops and yeast (Dutch: *gist*). Figure 1 shows the principal scheme of brewing. The first step is to malting (Dutch: *mouten*) barley which is converted into malt. In the next step, the mashing (Dutch: *maischen*), the malt is scraped (crushed), mixed with water and then reheated so that the released enzymes start working. The enzymes convert the starch known in the malt into sugars, which later it will be in alcohol. After filtering out impurities, one get's *wort*. The wort is then boiled with the addition of hops and possibly other extra flavorings. After the cooling, yeast is added. Yeast is a single-celled fungal species that converts the sugars present into alcohol. The fermentation lasts several days and can be done in different stages and circumstances, which is to a large extent the type of beer determine. After possible post-processing such as pasteurisation and adding extra carbon dioxide the beer can be put in bottles or barrels.

### 1.2 Costs and properties of the ingredients

There are three possibilities to obtain wort, as shown in the first three stages of the brewery process in Figure 1:

1. The purchase of malt extract and dilution with water.
2. Purchasing malt and mashing it yourself.
3. The purchase of barley and this itself malting and mashing.

If no distinction should be made between these three ingredients (barley, malt and malt extract), we will talk about "*malt sources*" in what follows. Although barley is cheaper than malt and malt is cheaper than malt extract, one have to take account of the process costs of malting and mashing. The process costs can each time be split up in a fixed cost (for, among other things, the maintenance and depreciation of the necessary machines and infrastructure) and a variable cost per kilogram (operating costs, personnel costs, etc.). If one intends to skip a certain process (eg leaving malt instead of malting barley), the corresponding process costs must of course not be worn (so also no fixed costs).



Figuur 1: Principal scheme of the brewery process (Dutch names).

### 1.3 Recipe and beer properties

#### 1.3.1 Converting between barley, malt, malt extract and wort

The beer we are going to brew has to meet certain properties as flavour, color and quality. In the following, we will define these properties based on the amounts of mout extract. For the conversion from barley to malt, you can use the following rules of thumb:

- Malting converts 1 kg of barley into 0.75 kg of malt.
- A certain wort composition based on 1 kg of malt per liter wort can also be achieved via 0.8 kg according to malt extract per liter of wort.

An important property of wort is the *SG-value* (“mass density”), expressed in kilograms per litre. The initial SG-value of the wort expresses how many sugar is present, en thus determines how many alcohol can be generated during fermentation. The following formula (thumb rule) expresses how many kilogram malt extract ( $m_{\text{tot}}$ ) is needed for  $V_{\text{wort}}$  litre wort with a certain SG-value:

$$\frac{m_{\text{tot}}}{V_{\text{wort}}} = 2.9 \cdot (SG - 1) \quad (1)$$

In this assignment, we will always assume that the initial SG value is 1,050 kg / l. We also assumes that every ingredient is available in unlimited quantities.

#### 1.3.2 The color of beer

An other quantity is the *EBC-value* of beer, malt or wort, who is expressed according to the European Brewery Convention (see Tabel 1). The EBC-value of a wort mixture (and the final beer) based on the several malt sources  $b$ , each in a amount  $m_b$  kilogram, corresponds to a weighted sum of the EBC-values of each component:

$$EBC_{\text{mixture}} = \frac{SG - 1}{0.0344} \sum_b EBC_b \cdot \frac{m_b}{m_{\text{tot}}} \quad (2)$$

Tabel 1: Beer colors with the accompanying EBC-value

| color       | EBC-value     |
|-------------|---------------|
| light blond | 6 to 8        |
| blond       | 9 to 12       |
| gold        | 13 to 19      |
| amber       | 20 to 29      |
| copper      | 30 to 45      |
| brown       | 46 to 75      |
| dark brown  | 76 tot 120    |
| black       | more than 120 |

An important detail: the values  $m_b$  in this formula (and all other formulas in this description) are amounts of malt extract. The amount of barley and malt must be first converted to the equivalent amount of malt extract before using these formulas.

### 1.3.3 The amount of hop and yeast

During the cooking of the wort, we add 1.3 grams of hops per litre of wort. You can assume that 5% of the volume evaporates, while cooking the wort. To the cooled liquid (the remaining one 95%) we add 75 grams of yeast per 100 litres. After the fermentation of this liquid is the beer ready. The added amounts of hops and yeast have no influence on the final volume of beer. Of course they have an influence on the properties of the beer.

### 1.3.4 The qualities of the ingredients

In this assignment, the ingredients are examined by a panel of beer experts and they earned a quality  $Q$ , going from 1 (weak) to 5 (excellent). The mean quality of a mixture is done for every ingredient group (malt sources, hops and yeasts respectively) separately. This is weighted mean. The weights are the mass fractions. For example, the mean quality of hop mixture, with several hops  $h$ , each with amount  $m_h$  and mean quality  $Q_h$ :

$$\frac{\sum_h m_h Q_h}{\sum_h m_h} \quad (3)$$

Don't forget to convert the amount of barley and malt to equivalent amount of malt extract if one has to determine the mean quality of the malt sources.

## 2 Practical issues

It is the goal to solve problem programmatic after modelling it as a linear problem. The project thus consists of two parts: one part for modelling the problem and one part for implementing the model.

### 2.1 Rules

- This project is individual. The assignment consists of this text and the data in the folder "Project" on Ufora. When a student helped another student, than both students have to report this in their paper. When non-reported collaboration is detected, this will have an influence in your grades.
- Send as soon as possible an email to [ljubomir.jovanov@ugent.be](mailto:ljubomir.jovanov@ugent.be) as a registration for the project.

- At the end of the project, you submit a paper with the solutions of the questions, the necessary explanations, tables, figures, etc. The paper must be submitted before 31 May. The paper and the code can exclusively be submitted digital; by uploading into the dropbox on Ufora or by mailing to [ljubomir.jovanov@ugent.be](mailto:ljubomir.jovanov@ugent.be).
- The paper must be a PDF file, where everything is nice organised: all questions are answered in the right order.
- The code has to be self-contained: nothing might be excluded to test your solution.

## 2.2 Implementation

For this project, you will have to solve a large optimisation problem. Because this is too difficult by hand, appropriate software will be used. This must be done in Matlab or Python. Because Matlab and Python have their own solver `linprog` for LP problems, you don't have to implement the simplex algorithm. The documentation can be found on <https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.linprog.html#scipy.optimize.linprog> for Python users and on <https://nl.mathworks.com/help/optim/ug/linprog.html> for Matlab users. Other, more advanced, methods to solve LP problems are not allowed.

A part of this project consists thus of reading the documentation. A good way to start experimenting with this function, is to apply them on several examples from the first chapters of the book of Vanderbei.

## 2.3 Dataset

Every student uses the same dataset with the properties (costs, qualities, etc) of the ingredients in processes. This dataset is simple file, `dataset_EN.txt`, that looks like this (“[...]” are omitted fragments):

```
# dataset dataset_EN.txt
# barley hops maltextracts maltprocess malts mashingprocess yeasts

# barley
# name EBC cost quality
barley01 65.0 0.101 3
barley02 124.0 0.11 3
barley03 116.0 0.107 5
[...]

# hops
# name cost quality
hop01 47.1 4
[...]

# maltextracts
# name EBC cost quality
maltextract01 72.0 9.57 3
[...]

# maltprocess
# name fixedcost variabelcost
malt proces 2410.0 0.39

# malts
# name EBC cost quality
```

```
malt01 134.0 2.83 3
[...]

# mashingprocess
# name fixedcost variabelekost
mashingprocess 1072.0 0.94

# yeasts
# name cost quality
yeast01 98.8 5
[...]
```

The lines that begins with “#” are guiding commentary, the other lines contain the actual data. The first line contains the name of the dataset. The second line contains the order of the ingredients and processes in this file. The next lines are the tables for each ingredient/process, separated by a empty line. The first line of every table is the type of the ingredient/process. The second line of the table contains the according fields of that type, separated by tabs. Then follow the data of the different species of that type, again separated by tabs. For the malt en mashing process, there is only one option.

The costs of the ingredients are in Euro per kg. The fixed costs of the processes are in. The variable cost of malting is in Euro per kg barley and the variable cost of mashing is in Euro per kg malt. The other values have no dimension.

Your program must be able to read files of this form. You don’t have to put to much effort in the flexibility of your reading procedure. You can assume that the order of the ingredients/processes is fixed en the order of the fields is fixed. The only thing that can change is the number of species of each ingredient (except for the two processes).

### 3 Assignment and questions

When asked to the solution of a linear program, don’t write the full vector of decision variables. Most of them will be zero. Just report the non-zero values, the corresponding decision variables/ingredients and also the properties (EBC-value, qualities) of the resulting beer.

#### 3.1 A first analysis: De cost-price as a function of the beer to be produced

1. How many litre beer can you produce from 25.44 kg barley? (If this does not seem “nice” you probably made a mistake).
2. As mentioned before, you can get wort in three different ways: based on barley, malt or malt extract. We consider the three different brewery processes whereby wort is obtained by one of these three. What is for each of these processes the minimal total cost of the beer production in function of the produced beer volume? For this question, you don’t have to take into account the beer properties (color, quality, etc). Show these three functions in a chart. Can you say something how the costs are spread between the different ingredient groups and the processes?

If the three malt sources can be used together, how will the minimal cost look than? Show this on your figure and explain.

#### 3.2 Formulate as a linear program

Consider the following optimisation problem

Minimise the total cost for brewing a certain volume  $V_{\text{beer}}$  beer with a certain color (determined by the EBC boundaries  $EBC_{\min}$  en  $EBC_{\max}$ ). The mean qualities of the different ingredient groups must be at least  $Q_{\text{malt}}$ ,  $Q_{\text{hop}}$  and  $Q_{\text{yeast}}$  respectively.

1. Formulate this as a mixed integer linear program (MILP). Be as strict as possible for the decision variables you incorporate ! Describe which decision variables you choose, how you obtain the cost function and the constraints.
2. Give a relaxation of this program.
3. Describe how you would solve this problem using brute force.
4. Write a program that models this problem en solves, based on the dataset you received. Make extra parameters for  $V_{\text{beer}}$ ,  $EBC_{\min}$ ,  $EBC_{\max}$ , etc. Solve the problem by using brute force. Make also an option to solve the relaxation of the MILP.

### 3.3 The production of 5000 litre amber beer

Consider the following optimisation problem:

Minimise the total cost for brewing 5000 litres of amber beer.

1. How does the MILP of this problem look? (compare with question 3.2 (1))
2. Solve the relaxation of this problem. What is the solution and the minimal cost?
3. It seems that most ingredients are not used. Why is this the case? Explain.
4. Which variables of the solution of the relaxation do not meet the requirements of the original problem? Try to round these variables zo get a feasible solution of the original problem. What is the cost and solution now?
5. Solve the original MILP. What is the solution and the cost?

### 3.4 Less or more?

Consider two small variations: instead of 5000 litres amber, one has to produce 50 litres amber beer and 25000 litres amber beer respectively.

1. Solve again the relaxation, the rounding, and the MILP (like in the previous question). What are the solutions and the corresponding costs?
2. Mark for each of the three beer volumes the three costs in the graph of question 3.1. Explain the position of the costs to the graph.
3. How does the cost per litre go? Does this meet your expectations?

### 3.5 Full problem

Consider now the full problem:

Minimise the cost of brewing 5000 litre amber beer, whereby the mean qualities of the respective ingredient groups must be at least 3.

Solve this problem with your program. What is the solution and the cost? Compare with the results from Question 3.3.

**3.6 From light blond to black**

Solve the full problem like in Question 3.5, but now for every color in Table 1.

1. Visualise the cost price in function of the beer color.
2. Explain why certain beer colors are the cheapest/most expensive.
3. Are all colors reachable? Why (not)?

**3.7 Maximise the quality**

We turn the problem around:

Maximise the minimum of the three mean qualities. The cost price is unlimited, but it must be 5000 litre amber beer.

Formulate this as a mixed integer linear program and describe how you obtained it. You don't have to solve this problem with your program. Can the minimum of the mean quality be 5? Explain.